



Sony's \$70,000 98-inch 8K TV

Sony's upcoming flagship TV, while cheaper than the Q900, is still pretty expensive. <https://digit.in/may19-7>



Tesla robotaxi in 2020

Tesla to launch robotaxis as part of broader vision for an autonomous ride-sharing by 2020. <https://digit.in/may19-8>



A fool's errand

Agent001 remains skeptical about high-resolution displays for gaming

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At a recent event, I was engaged in a pretty decent conversation regarding our personal rigs with a friend of mine. We'd both recently upgraded and had both gone down different paths. I'd decided to stick with 1080p 144 Hz high frame-rate display whereas my friend had decided with a 4K 60 Hz display which he ran at 2K since he was experiencing extremely low frame rates at 4K resolution. We couldn't give up an opportunity like this to test different games under different settings, so we started out with a few well-known titles which were optimised for low-end machines. Both of us had the same GTX 1080 Ti graphics card so there was barely any difference between both our rigs.

There are a few simple facts about games and the frame rates you receive. The moment you bump up the graphics settings to max, the frame rate will drop. Increasing the resolution will cause the frame rate to drop. Games are primarily GPU bound but a lot of the current gen games are increasingly veering towards the CPU being the bottleneck. What we observed was pretty much what we had been seeing for the past few years. Higher resolutions are simply not optimal for heavy applications such as gaming.

If you take your graphics card, the first thing you need to work out is whether it can crunch all the frames quickly. Game developers often look at 60 FPS as the optimal frame-rate to maintain to ensure a good experience. Maintaining 60 FPS at higher resolutions isn't easy but you can make do at 1080p. Every new gen-

ers for quite some time now but even game developers would rather spend more time optimising the graphics than work on scaling the game engine with respect to the CPU. We now have HEDT CPUs with 32 cores but a video game will hardly make use of all those cores. Heck, even if you drop the graphics settings, it'll still make no

difference. A game running at minimal settings on a Quad-core CPU will give just about the same frame rates as a game running on an Octa-core CPU. In most of our benchmarks, we find performance to be greater on Quad-core CPUs than any other configuration.

CPU cores are capable of running a myriad variation of calculations compared to GPU cores, which is why it is easier



Higher resolutions aren't always a good thing

to squeeze in more of the dumber GPU cores in the same physical area compared to CPU cores. Think of the CPU cores as a massive maze and the GPU cores as a much simpler maze. Because the calculations pertaining to one video frame have to spend a lot more time navigating a CPU core, the number of frames it can process in one second is limited. This is why, if you take a modern video game and completely reduce the graphics

eration of graphics cards add more streaming multiprocessors and the performance scales remarkably well, even for older games.

However, if you take a look at the CPU, then it has remained in the same state for the last decade or so. Video game engines are optimised to make the most of 4 cores and whatever more you throw at it gives you no worthwhile returns. CPU manufacturers have been working with game develop-